PRACTICAL APPLICATION:
2. Soar playbook development:

Playbook Creation:

Attempted to install Splunk Phantom on multiple environments, including CentOS and Ubuntu via VirtualBox. However, the installation was unsuccessful due to system limitations. Configuration steps could not be completed as the system did not meet the minimum requirements of memory, RAM and resource capacity. Multiple online resources and youtube tutorials were referred to for guidance, but the installation still failed. As a result, Phantom could not be deployed for testing in the current setup.

Description of the Playbook:
The playbook is made to automatically respond to phishing alerts from Wazuh. When an alert comes in, the system first checks the source IP against threat intelligence. If the IP is found to be harmful, it is blocked automatically using CrowdSec to stop further attacks. At the same time, an incident ticket is created in TheHive with all alert details for record and investigation. The playbook also keeps a log of all actions for auditing and makes sure that response is fast, reducing the need for manual work. By connecting detection, blocking, and incident management, the playbook helps quickly and safely handle phishing threats.

Using draw.io created a flowchart for visual representation of the playbook. It shows how phishing alerts from Wazuh are handled step by step—from checking the IP, to blocking malicious sources with CrowdSec, and finally creating an incident ticket in TheHive. This visual representation helps to easily understand the workflow and the automated response process.”

Flowchart:

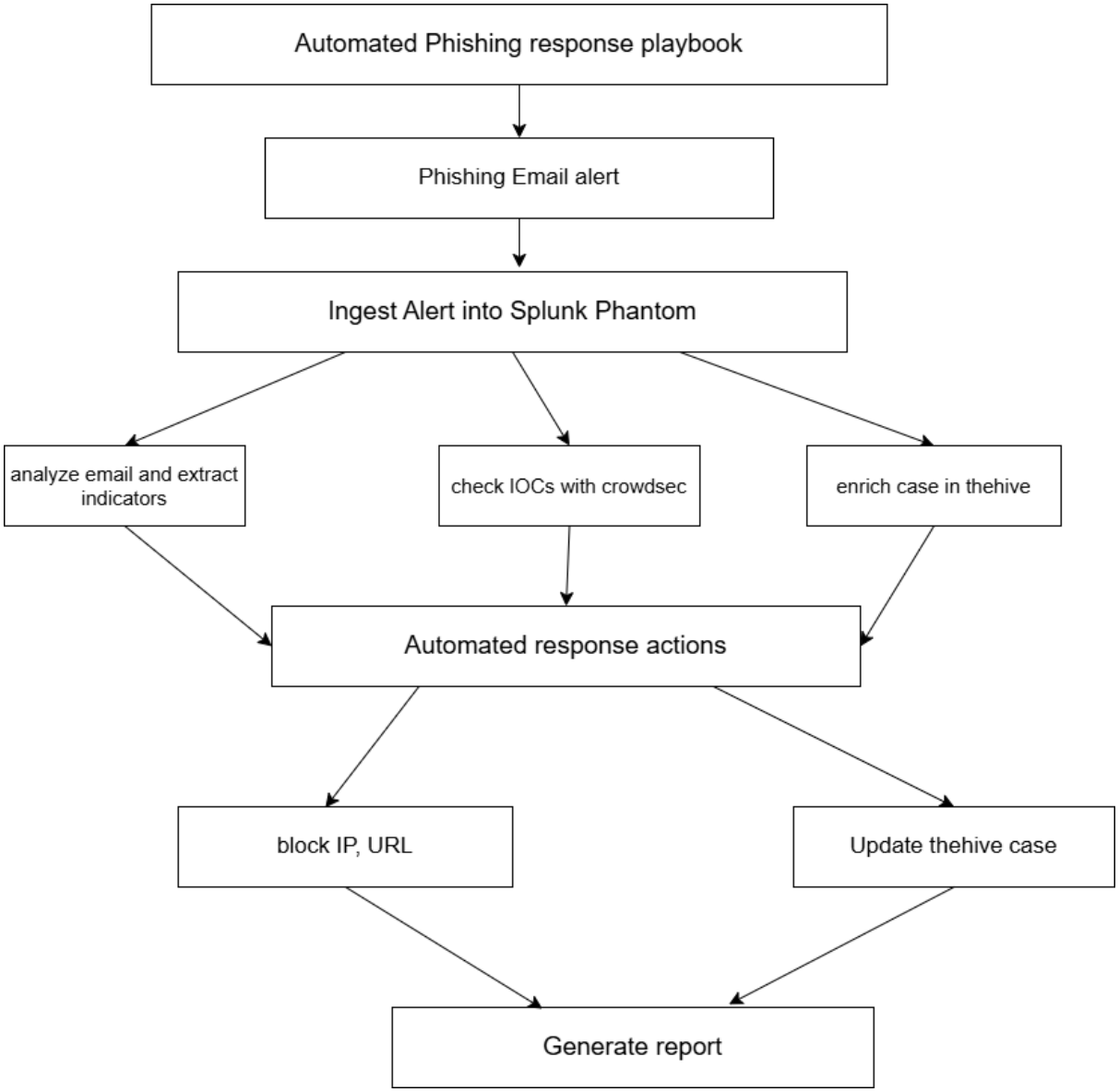

Figure 1: Conceptual Flowchart of Automated Phishing Response Playbook using Splunk Phantom, CrowdSec, and TheHive

Playbook Test:
Simulated a phishing alert in Wazuh and documented it :

On wazuh Wazuh manager server, created a test rule

```
sudo nano /var/ossec/etc/rules/local_rules.xml
```

and add the rules as shown in below

```
  GNU nano 7.2                /var/ossec/etc/rules/local_rules.xml
    <mitre>
      <id>T1583</id>
    </mitre>
  </rule>
</group>

<!-- Mock Suspicious File Download Detection -->
<group name="local,mock_alerts">

  <rule id="100005" level="10">
<!--    <program_name>logger</program_name> -->
    <match>suspicious-file.exe</match>
    <description>Mock alert: Suspicious File Download</description>
    <mitre>
      <id>T1204</id> <!-- User Execution: Malicious File Download -->
    </mitre>
  </rule>

</group>

<!-- Phishing Email Detection -->
<group name="local,phishing_test">

  <rule id="100600" level="12">
    <if_group>syslog</if_group>
    <match>Phishing email received</match>
    <description>Phishing email detected: Spearphishing T1566</description>
    <mitre>
      <id>T1566</id>
    </mitre>
  </rule>

</group>
```

Figure 2: created a rule

Then restart wazuh manager

```
sudo systemctl restart wazuh-manager
```

after that send a test log through logger

```
logger "Phishing email received from 192.168.1.102"
```

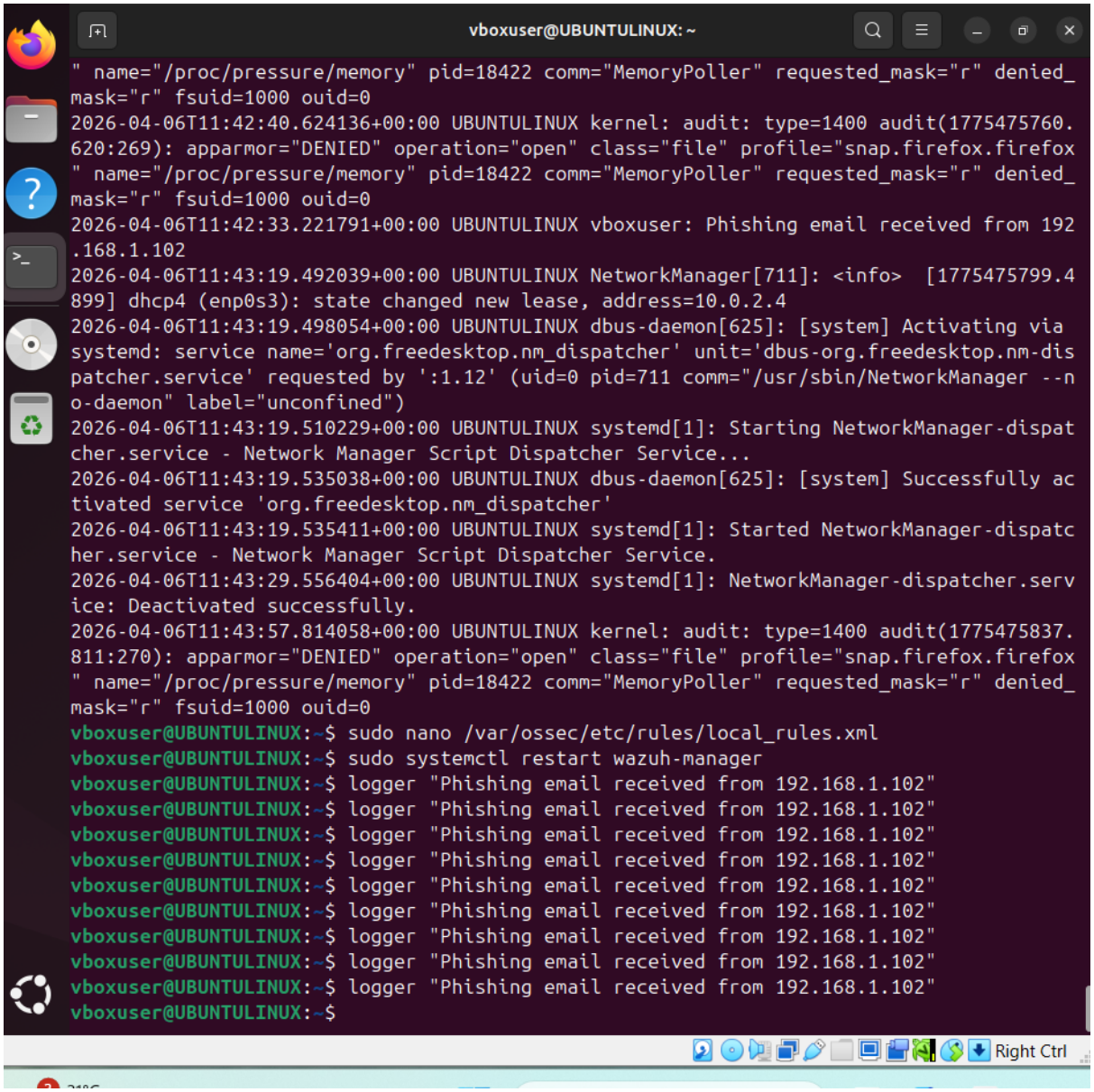

Figure 3: restarting wazuh manager and send test log through logger

In wazuh dashboard detect a alert,
In search bar type

rule.id: "100600"

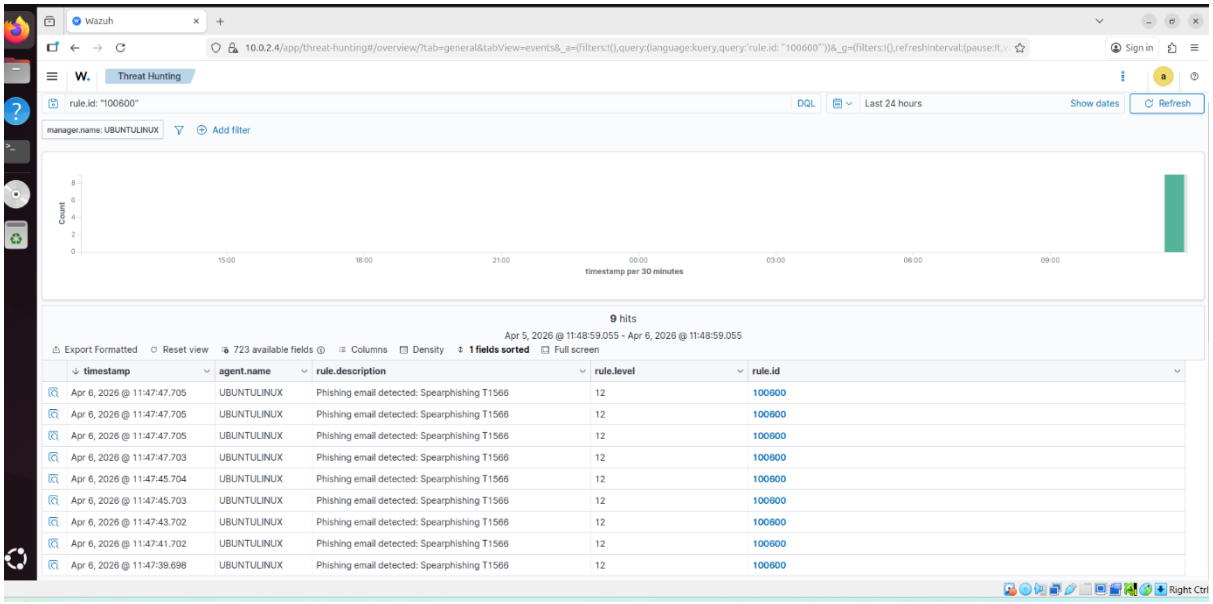

Figure 4: alerts detected in wazuh

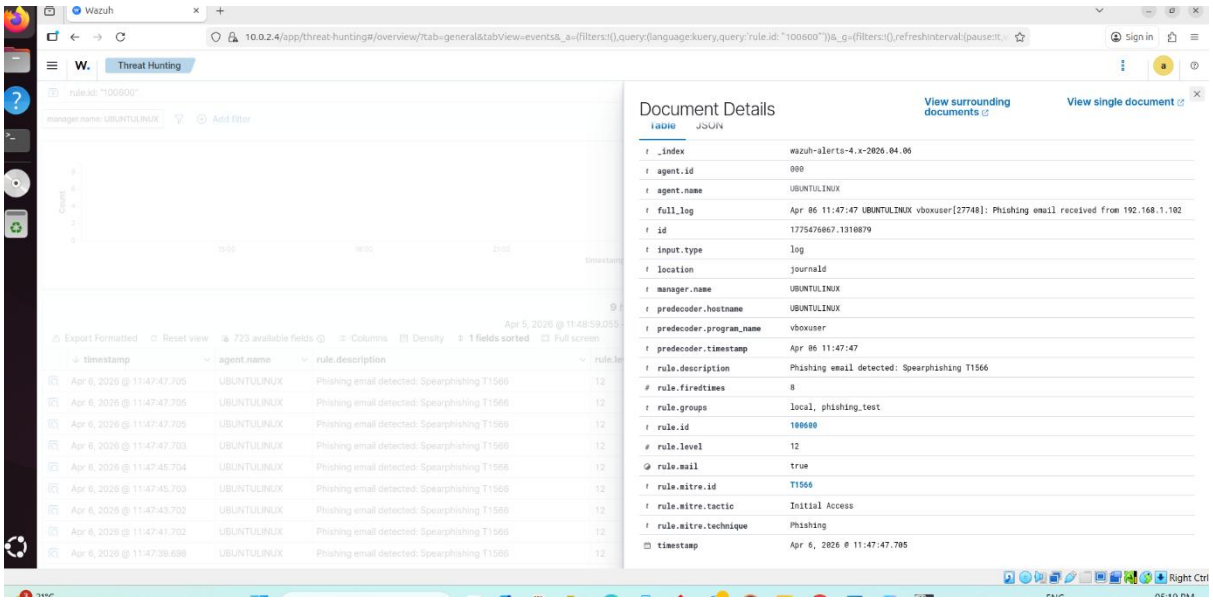

Figure 5: alert details

Check ip in virus total :

Goto virus total and search this ip found in alert – 192.168.1.102

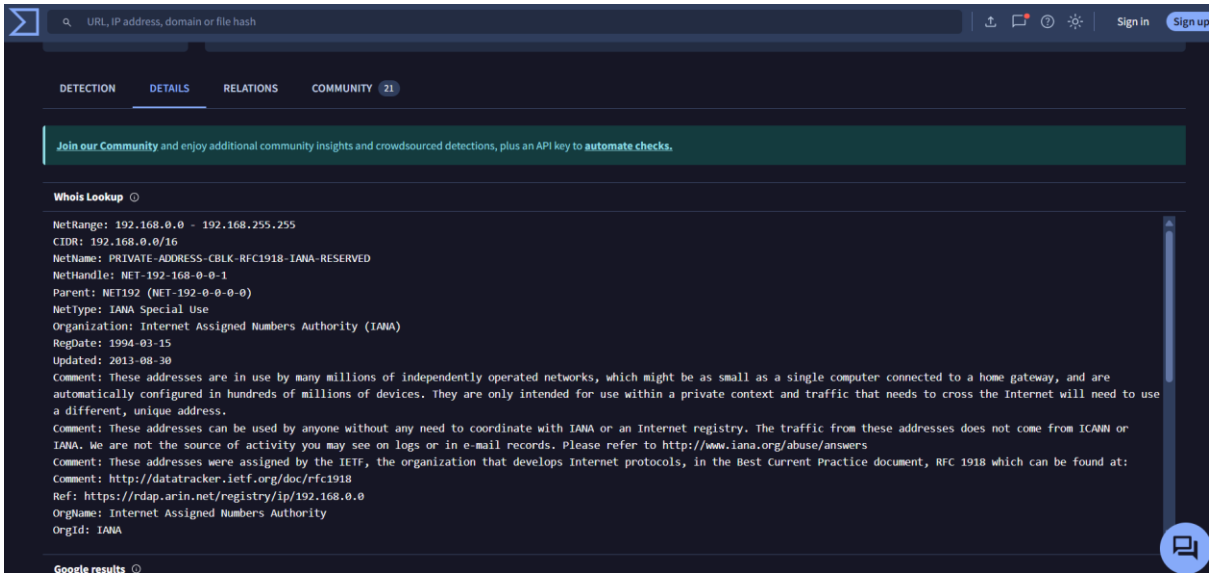

Figure 6

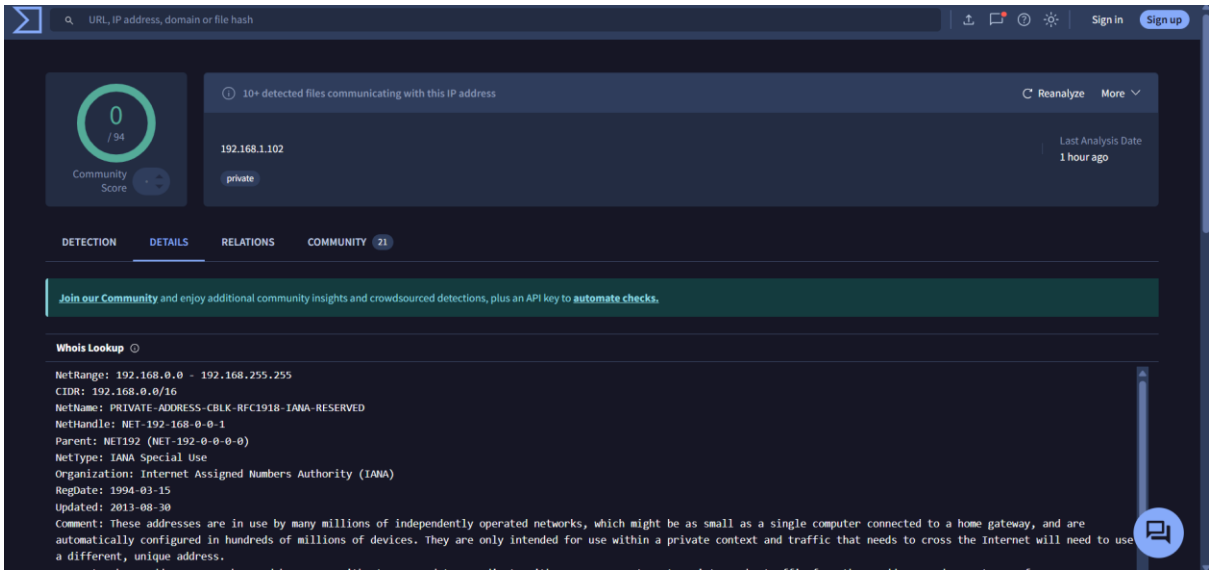

Figure 6

Figure 5 and 6 is the output from virus total

The IP address 192.168.1.102 is a private address within the 192.168.0.0–192.168.255.255 range, reserved for internal network use and not routable on the public internet. VirusTotal analysis shows a 0/94 detection score, indicating that no security vendors flagged this IP as malicious. However, multiple files have communicated with this IP, suggesting normal internal network activity or potential internal lateral movement. While the IP itself is not malicious, it is recommended to review the files and internal traffic logs to ensure that there is no unauthorized or suspicious activity within the network.

Block the IP manually:

Using crowdsec block the ip manually,

sudo cscli decisions add --ip 192.168.1.102 --reason "Blocked due to phishing alert"

verify by,

sudo cscli decisions list

Figure 7: blocked IP using crowdsec

Document:

Playbook step	status	notes
Check ip	success	Checked ip in virustotal
Block ip	success	Using crowdsec blocked ip 192.168.1.102

Summary:

Due to system limitations, Splunk Phantom could not be successfully installed and configured in the current lab environment. Multiple installation attempts were made on both CentOS and Ubuntu virtual machines using VirtualBox, but the setup failed because the system did not meet the minimum hardware and resource requirements, especially in terms of RAM, memory, and overall system capacity. Several online resources and YouTube tutorials were referred to during the installation process, but deployment was still unsuccessful. As an alternative, the playbook workflow was demonstrated conceptually using a flowchart and tested manually. A phishing alert was simulated and detected in Wazuh, the suspicious IP was checked in VirusTotal, and the malicious IP was blocked successfully using CrowdSec. This manual approach helped validate the core incident response steps of the intended SOAR playbook.

A manual simulation of the phishing response playbook was carried out to validate the workflow. First, a phishing alert was triggered and detected in Wazuh using a custom test rule. After the alert was generated, the suspicious source IP was identified and checked in VirusTotal to verify whether it was malicious. The IP was

then blocked manually using CrowdSec to simulate containment of the threat. In addition, a flowchart was created to visually represent the playbook process, showing the sequence of alert detection, IP verification, and blocking action. This manual test confirmed that the main steps of the phishing incident response workflow were successfully performed.